

Memory Hierarchy

The memory hierarchy in computer architecture refers to the organization of different types of memory storage in a system, arranged in hierarchy based on their speed, size and cost.

⇒ The primary goal is to provide a balance between access speed and storage capacity.

Types

⇒ The hierarchy of memory is mainly divided into two types:

- External or Secondary memory
- Internal Memory: or primary Memory

External or Secondary memory:

⇒ It consists of Magnetic Tape, optical disk, Magnetic disk, i.e. It includes peripheral storage devices that are accessible by the system processor.

via I/O module.

External memory or Primary

- It consists of the CPU registers, Cache Memory and Main memory. It is accessible directly by the processor.

1) Registers:

Location:

- Fastest, directly within the CPU.

Purpose:

- Used for storing small amounts of data that the CPU is actively processing.

Size:

- Very limited, typically measured in bytes.

2) Cache memory:

Location:

- Situated between registers and RAM. Also known as ~~an~~

Purpose:

- Temporarily stores frequently accessed or recently used data and instructions to speed up CPU operations.

Size:

- Relatively small compared to RAM but larger than registers.

~~8KB to 16KB~~

- KB to MB

3) RAM:

Location:

- Also known as main memory. Store data that is currently used by CPU.

- Slower than Cache but faster than secondary storage, usually on the

motherboard.

Purpose:

- Holds data and instructions that are actively being used or processed by the CPU.

Size:

- Larger than cache but smaller than secondary storage.
4GB/8GB to 16GB

Static RAM:

- Stores the binary information in flip flops. Remains valid until power is supplied.

Dynamic RAM:

- Stores binary information as a charge on the capacitor.
- Requires refreshing circuitry.

Secondary device:

Hard drives / SSDs

Location:

non-volatile storage located outside the CPU

Purpose:

- Stores data and programs for long term use even when the power is off.

- Store data and instructions that are not currently in use by the CPU.

Speed:

- Slower access speed compared to RAM

Size:

- Much larger capacity and least expensive.

5. Magnetic Disk:

- Simple circular plates that are fabricated with either a metal or a plastic or a magnetized material.
- Work at high speed inside the computer.

6) Magnetic Tape:

- Simply a magnetic recording medium that is covered with a plastic film.
- Commonly used for the backup of data.
- Requires large amount of time for accessing the data.

7) Tertiary storage (Backup drives, cloud storage):

Location:

- External to the computer system.

Purpose:

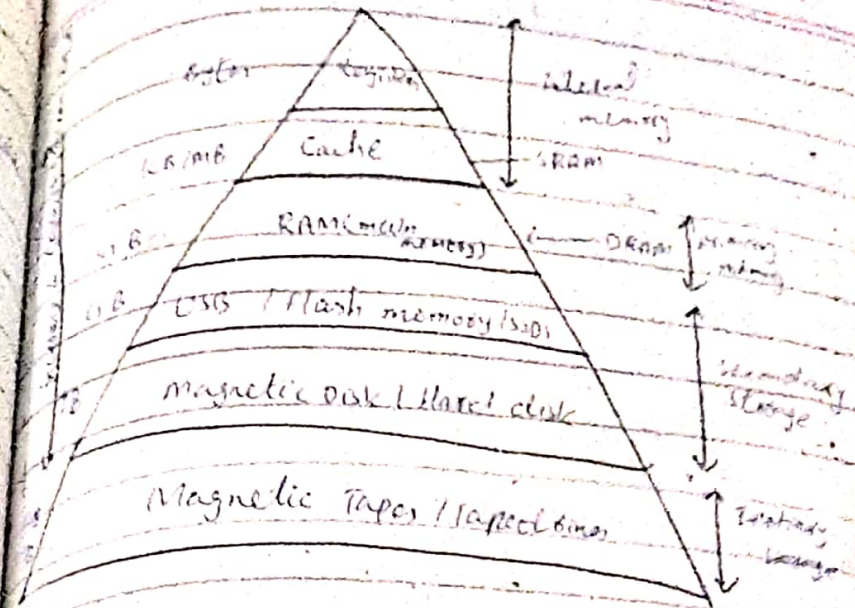
- Used for back ups and long term archiving.

Size:

- Typically larger capacity for archival purposes.

Speed:

- Much slower access compared to secondary storage.



Access Patterns:

Temporal Locality: Accessing the same memory locations repeatedly in a short time.

Spatial Locality:

Accessing memory locations that are near each other.

Caching Techniques:

Cache Lines

- Blocks of contiguous memory that are loaded into the cache together.

Associativity:

- Determines how many cache lines a particular set in the cache can hold.

Characteristics of memory hierarchy:

* Speed

⇒ Memory hierarchy is organized based on access speed, which faster types close to the CPU for quicker access to frequently used data.

* Size

⇒ memory types in the hierarchy vary in size, with registers being the smallest and tertiary storage having the largest capacity.

* Cost:

⇒ The cost of memory increases closer to the faster, smaller, and more volatile types of memory. Cache and registers, being faster but more expensive.

* Volatility

- Registers, cache and RAM, are volatile.
- Secondary and tertiary storage are non-volatile.

* Hierarchy Levels:

⇒ The memory hierarchy is structured in multiple levels, typically including registers, cache (L_1, L_2, L_3), RAM, secondary storage (HDD, SSD) and tertiary storage (external drives, cloud storage).

⇒ These characteristics collectively contribute to the effectiveness of the memory hierarchy in providing a balance between speed, capacity, and cost, optimizing overall system performance.

Advantages:

- Speed and Responsiveness
- Efficient resource utilization
- Power efficiency
- help in managing the memory in better way.

Purpose of MH:

⇒ The memory hierarchy aims to optimize the trade-off between speed and capacity, ensuring that the CPU can effectively access the most relevant data while minimizing the impact of slower memory types.

System Supported memory standards:

Level	1	3	4
Registers	Cache	Main memory	Secondary memory
Size	< 1KB	< 16MB	< 16GB
Implementation	On-chip / SRAM	DRAM (Capacitor memory)	Magnetic
Access time	0.5 ns to 0.5 ns	8 ns to 25 ns	50 ns to 100 ns
Bandwidth	20000 to 1 lakh MB	5000 to 15000 MB	1000 to 5000 MB
Managed by	Compiler	Hardware	OS
Backing mechanism	from main cache	from main memory	from secondary memory